Shape and Space 3 — The tangent ratio · Angles of elevation and depression · Bearings

Calculator allowed. Give answers correct to 3 significant figures. Draw a labelled right-angled triangle for every question. Marks in brackets. ★ = harder question.

1 In a right-angled triangle, the side adjacent to a $38°$ angle is 14 cm long. Find the length of the side opposite the $38°$ angle. [3]

2 An aeroplane takes off and climbs in a straight line. When it has travelled 600 m horizontally it is 250 m above the ground. Calculate its angle of elevation θ from the take-off point. [3]

3 From the top of a lighthouse, the angle of depression of a small boat is $18°$. The boat is 150 m from the base of the lighthouse. Find the height of the lighthouse. [4]

4 A hot-air balloon rises vertically from a point on the ground. An observer stands 280 m away. When the balloon stops for the first time, its angle of elevation from the observer is $30°$; when it stops at its highest point the angle is $55°$. [7]

a Find the height h_1 of the first stop and the further height h_2 it then rises.

b The balloon takes 1 minute to reach the first stop and a further 6 minutes to reach the top. Calculate the average speed of the whole ascent in m/s.

5 A ship sails 8 km due north from a port P, then 6 km due east to a point Q. [8]

a Draw a diagram of the route.

b How far is Q from P in a straight line?

c Calculate the bearing of Q from P.

d Write down the bearing of P from Q.

6 A vertical mast stands on level ground. From a point 40 m from its base, the angle of elevation of the top is $35°$. From a point 25 m further away, in the same straight line, find the angle of elevation of the top. [4]

Total: 29 marks

Check: $\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$ — label the sides from the angle you're using · angle of depression = angle of elevation from the other end · bearings: from north, clockwise, three figures · reverse bearing: ± 180°.